

PUMaC 2017 Power Round Solutions

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The title of the round, "I Ain't Even Lie-in'...", was inspired by "Dunked On" by Froggy Fresh.

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1 Linear Algebra I (20 points)

1.1 Vector Spaces (5 points)

Problem 1.1.1. (2 points) Show that any linearly independent set of size $\dim V < \infty$ in the vector space V over the field F is a basis of V .

Solution: Suppose not, and $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\} \subset V$. Add in $\mathbf{v}_{n+1} \in V \setminus \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$; we will show that $\{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{v}_{n+1}\}$ is linearly dependent.

Give V the basis $\mathbf{w}_1, \dots, \mathbf{w}_n$, which we know exists by the definition of dimension. Since $V = \text{span}\{\mathbf{w}_i\}$ we can write $\mathbf{v}_1 = a_1\mathbf{w}_1 + \dots + a_n\mathbf{w}_n$ where, without loss of generality, $a_1 \neq 0$. So we can rearrange and then we find that we can replace $\mathbf{w}_1$ with $\mathbf{v}_1$ in the basis. We repeat with each of the other elements until we find that in fact $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ are sufficient to span.

Problem 1.1.2. (3 points) Show that $\mathbb{R}[x]$ is a real vector space, and that $\mathbb{C}[x]$ is a complex vector space (where $F[x]$ is the set of finite-degree polynomials with coefficients in F). Is $\mathbb{R}[x]$ a complex vector space? Is $\mathbb{C}[x]$ a real vector space?

Solution: First part: Adding polynomials over a given space is clearly valid, since it will just get a different polynomial, also with coefficients in the field (since the field is closed). Second part: no; $1 \in \mathbb{R}[x]$ but $i \cdot 1 \notin \mathbb{R}[x]$, making it not closed under complex scalar multiplication. Third part: yes, since a complex number times a complex number is still complex.

1.2 Linear Mappings, Inner Products (11 points)

Problem 1.2.1. (2 points each) Fix $N \in \mathbb{N}$. Let $a(x)$ take the form $a(x) = \sum_{n=0}^N a_n x^n$ and let F_N be the set of degree- N polynomials in $F[x]$. For each of the following functions $f_i : F_N \rightarrow F[x]$, determine with proof whether or not it is a linear map.

- (i) $f_1 : a(x) \mapsto a(x) + a(1)$
- (ii) $f_2 : a(x) \mapsto a(x) + a(1)x^{N+1}$
- (iii) $f_3 : a(x) \mapsto \sum_{n=0}^N (a_n - a_{N-n})x^n$

Solution:

- (i) Yes; $f_1((\alpha a + \beta b)(x)) = (\alpha a + \beta b)(x) + (\alpha a + \beta b)(1) = \alpha a(x) + \beta b(x) + \alpha a(1) + \beta b(1) = \alpha f_1(a(x)) + \beta f_1(b(x))$ (where $\alpha, \beta \in F; a, b \in F[x]$).
- (ii) Yes; $f_2((\alpha a + \beta b)(x)) = \alpha a(x) + \beta b(x) + (\alpha a(1) + \beta b(1))x^{N+1} = \alpha(a(x) + a(1)x^{N+1}) + \beta(b(x) + b(1)x^{N+1}) = \alpha f_2(a(x)) + \beta f_2(b(x))$ (where $\alpha, \beta \in F; a, b \in F[x]$).
- (iii) Yes; $f_3((\alpha a + \beta b)(x)) = \sum_{n=0}^N (\alpha a_n + \beta b_n - \alpha a_{N-n} - \beta b_{N-n})x^n = \alpha f_3(a(x)) + \beta f_3(b(x))$ (where $\alpha, \beta \in F; a, b \in F[x]$).

Problem 1.2.2. (3 points) Describe as precisely as possible all possible inner products on $\mathbb{C}^2$.

Solution: Using the properties in Definition 1.2.F, we find that

$$\langle (c_1, c_2), (c_3, c_4) \rangle = c_1 \overline{c_3} \langle (1, 0), (1, 0) \rangle + c_1 \overline{c_4} \langle (1, 0), (0, 1) \rangle + c_2 \overline{c_3} \langle (0, 1), (1, 0) \rangle + c_2 \overline{c_4} \langle (0, 1), (0, 1) \rangle.$$

We now must simply determine the inner product of each pair of basis vectors $\langle \mathbf{e}_i, \mathbf{e}_j \rangle$ where $\mathbf{e}_i = (2 - i, i - 1)$. We know that $\langle \mathbf{e}_i, \mathbf{e}_i \rangle \in \mathbb{R}^+$ by the first and fourth properties, so we can

call $r_i = \langle \mathbf{e}_i, \mathbf{e}_i \rangle$. We know that $\langle \mathbf{e}_1, \mathbf{e}_2 \rangle = \overline{\langle \mathbf{e}_2, \mathbf{e}_1 \rangle}$ so we can let $0 \neq c = \langle \mathbf{e}_1, \mathbf{e}_2 \rangle$. Then we get the inner product

$$\langle (c_1, c_2), (c_3, c_4) \rangle = c_1 \overline{c_3} r_1 + c_1 \overline{c_4} c + c_2 \overline{c_3} c + c_2 \overline{c_4} r_2$$

for any $r_1, r_2 \in \mathbb{R}^+$ and $c \in \mathbb{C} \setminus \{0\}$, which can be verified by checking the properties in Definition 1.2.F.

Problem 1.2.3. (2 points) If ϕ is an isomorphism, show that ϕ^{-1} is also an isomorphism.

Solution: As it is a bijective homomorphism, ϕ has its ϕ^{-1} also a homomorphism that is 1-to-1. Thus, ϕ^{-1} is a homomorphism where its inverse is well-defined (since it is 1-to-1) and a homomorphism, making it also a bijective homomorphism.

1.3 Matrices (4 points)

Problem 1.3.1. (1 point) Does matrix multiplication associate? Why or why not? Answer the same question about commutativity.

Solution: First question: application of linear maps associates. Second question: no, $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$ and $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

Problem 1.3.2. (1 point) Show that, for linear mappings $T_1, T_2 : \mathbb{R}^m \rightarrow \mathbb{R}^n$ represented by the $n \times m$ matrices $\{a_{i,j}\}$ and $\{b_{i,j}\}$, respectively, the linear mapping $T_1 + T_2 : \mathbb{R}^m \rightarrow \mathbb{R}^n$ can be represented by the $n \times m$ matrix $\{a_{i,j} + b_{i,j}\}$ (where in each instance i ranges along 1 to n and j from 1 to m).

Solution: Give $\mathbb{R}^m$ the basis $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ and $\mathbb{R}^n$ the basis $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$. We'll write the action of T_1 as sending $\mathbf{v}_k \mapsto a_{1,k}\mathbf{w}_1 + \dots + a_{n,k}\mathbf{w}_n$ and of $T_2 : \mathbf{v}_k \mapsto b_{1,k}\mathbf{w}_1 + \dots + b_{n,k}\mathbf{w}_n$. Adding these maps vector-by-vector can be shown to compare directly with what the sum of the matrices looks like.

Problem 1.3.3. (2 points) Show that, for matrices $T_1 : \mathbb{R}^n \rightarrow \mathbb{R}^p$ and $T_2 : \mathbb{R}^m \rightarrow \mathbb{R}^n$ and vector $\mathbf{v} \in \mathbb{R}^m$, $T_1(T_2\mathbf{v}) = (T_1 \times T_2)\mathbf{v}$.

Solution: Give $\mathbb{R}^m$ the basis $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$, $\mathbb{R}^n$ the basis $\{\mathbf{w}_1, \dots, \mathbf{w}_n\}$, and $\mathbb{R}^p$ the basis $\{\mathbf{x}_1, \dots, \mathbf{x}_p\}$. We'll write the action of T_1 as sending $\mathbf{v}_k \mapsto a_{1,k}\mathbf{w}_1 + \dots + a_{n,k}\mathbf{w}_n$ and of $T_2 : \mathbf{w}_k \mapsto b_{1,k}\mathbf{x}_1 + \dots + b_{p,k}\mathbf{x}_p$. Composing these maps vector-by-vector can be shown to compare directly with what the product of the matrices looks like.

2 Linear Algebra II (22 points)

2.1 Eigen- (7 points)

Problem 2.1.1. (1 point) Show that each eigenspace V_λ is a subspace of V .

Solution: If $v, w \in V_\lambda$ then $M(av + bw) = aMv + bMw = a\lambda v + b\lambda w = \lambda(av + bw)$, as desired.

Problem 2.1.2. (1 point) Why does any mapping of a finite-dimensional space have a finite number of distinct eigenvalues?

Solution: Suppose not, and there were an infinite number. Then in particular we can choose more eigenvectors corresponding to distinct eigenvalues than the dimension of the space. These eigenvectors are linearly independent since otherwise they would all belong to the same eigenspace, so this contradicts the definition of dimension.

Problem 2.1.3. (2 points) Suppose $M : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear map with matrix $\begin{bmatrix} 2 & 0 \\ 1 & 7 \end{bmatrix}$. Find each eigenvector and the corresponding eigenvalue of M . (Hint: which linear map sends the vector $\mathbf{v}$ to $c\mathbf{v}$ ($c \in \mathbb{R}$)?)

Solution: $M \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2x \\ x + 7y \end{bmatrix} = \begin{bmatrix} cx \\ cy \end{bmatrix}$. We immediately find 2 cases: $c = 2$ or $x = 0$.

In the former case, $x + 7y = 2y \implies x = -5y$. We thus get the 2-eigenvector $\begin{bmatrix} -5t \\ t \end{bmatrix}$ for $t \in \mathbb{R}$. If $x = 0$ then $7y = cy$. $y = 0$ is not an option (this would just generate the trivial eigenspace) so we find $c = 7$ and the 7-eigenspace is generated by $\begin{bmatrix} 0 \\ t \end{bmatrix}$ for $t \in \mathbb{R}$.

Problem 2.1.4. (3 points) For each function in Problem 1.2.1, if it is a linear map, determine its eigenvalues and classify as best you can the associated eigenvectors.

Solution:

- (i) If $a(x) + a(1) = \lambda a(x)$ then $(\lambda - 1)a(x) = a(1)$ for all x . This means that either $a(x)$ is a constant or $\lambda - 1 = 0$. In the former case, if $a(x)$ is a nonzero constant then we get $\lambda = 2$, and $V_2 = F$. In the latter case, we get $a(1) = 0$, so the V_1 is everything in $F[x]$ with their sum of coefficients equal to zero.
- (ii) $a(x) + a(1)x^{N+1} = \lambda a(x)$ only when $a(1) = 0$, because x^{N+1} does not appear on the right-hand side. Thus, we get that V_1 is everything in $F[x]$ with their sum of coefficients equal to zero.
- (iii) Let $\tilde{a}(x) = \sum a_{N-n}x^n$. Then in order to have $a(x) - \tilde{a}(x) = \lambda a(x)$ we need $\tilde{a}(x) = (1 - \lambda)a(x)$; this means $a_{N-n} = (1 - \lambda)a_n = (1 - \lambda)^2 a_{N-n}$; the only solutions are $\lambda = 0, 2$. V_0 is the polynomials that are symmetric; V_2 is the polynomials that are antisymmetric (i.e. $a_n = -a_{N-n}$).

2.2 Trace (12 points)

Problem 2.2.1. (2 points) What is $\dim \mathfrak{sl}_n$? Find a basis.

We want to have trace zero, so we need to have $n - 1$ degrees of freedom on the main diagonal. The remaining $n^2 - n$ entries not on the main diagonal can be anything, so there are $n^2 - n$ extra degrees of freedom. The final degree is $n^2 - 1$. A basis could be

$\begin{bmatrix} \ddots & & & \\ & 1 & & \\ & & \ddots & \\ & & & -1 \end{bmatrix}$ where there are some number of zeroes along the first $\ddots$ and $n - 2$

minus that number along the second $\ddots$, and zeroes elsewhere. The other basis elements could have a 1 at row i and column j , one for each i and j ($i \neq j$).

Problem 2.2.2. (10 points) Classify as best you can the elements of $\mathfrak{so}_n$ and $\mathfrak{sp}_n$; find $\dim \mathfrak{so}_n$ and $\dim \mathfrak{sp}_n$.

Solution:

$\mathfrak{so}_{2n+1}$: Elements here take the form $\begin{bmatrix} 0 & C^T & -B^T \\ B & M & P \\ -C & Q & -M^T \end{bmatrix}$ where B and C each have n rows

and 1 column, and P and Q both satisfy $X = -X^T$. P and Q have zeroes along the diagonal and total freedom in the upper triangle, for $2 \cdot \frac{(n-1)n}{2}$ degrees of freedom. M

has n^2 degrees of freedom. B and C each have n degrees of freedom. Thus the total dimension is $2n^2 + n$.

$\mathfrak{so}_{2n}$: Elements here take the form $\begin{bmatrix} M & P \\ Q & -M^T \end{bmatrix}$ where P and Q both satisfy $X = -X^T$. P and Q have zeroes along the diagonal and total freedom in the upper triangle, for $2 \cdot \frac{(n-1)n}{2}$ degrees of freedom. M has n^2 degrees of freedom. Thus the total dimension is $2n^2 - n$.

$\mathfrak{sp}_{2n}$: Elements here take the form $\begin{bmatrix} M & P \\ Q & -M^T \end{bmatrix}$ where P and Q both satisfy $X = X^T$. P and Q have total freedom in the upper triangle that includes the main diagonal, for $2 \cdot \frac{n(n+1)}{2}$ degrees of freedom. M has n^2 degrees of freedom. Thus the total dimension is $2n^2 + n$.

2.3 Semisimplicity & nilpotency (3 points)

Problem 2.3.1. (3 points) Show that if X and Y are matrices that commute and X is nilpotent then $\text{tr}(XY) = 0$.

Solution: We note that if $X^k = 0$ then $(XY)^k = X^k Y^k = 0$, so XY is also nilpotent. It then just remains to be shown that all nilpotent matrices have trace zero. We know that $X^n \mathbf{v} = \lambda^n \mathbf{v}$ when $\mathbf{v}$ is a λ -eigenvector. However, setting $n = k$ gives $\lambda = 0$. Since trace is sum of eigenvalues, we are done.

3 Lie Algebras I (77 points)

3.1 What is a Lie algebra? (26 points)

Problem 3.1.1. (1 point) Let $[f, g] = f \circ g - g \circ f$ for $f, g \in \mathfrak{gl}_n$. Show that the Jacobi identity holds for $f, g, h \in \mathfrak{gl}_n$, and therefore that $\mathfrak{gl}_n$ is a Lie algebra.

Solution: $[f, [g, h]] = [f, gh - hg] = fgh - fhg - ghf + hgf$. $[g, [h, f]] = [g, hf - fh] = ghf - gfh - hfg + fhg$. $[h, [f, g]] = [h, fg - gf] = hfg - hgf - fgh + gfh$. Adding gets cancellation to 0.

Problem 3.1.2. (1 point) Explain why two Lie algebras of different dimension cannot be isomorphic.

Solution: Their underlying vector spaces must be isomorphic.

Problem 3.1.3. (4 points) Show that if L_1 and L_2 are abelian Lie algebras, they are isomorphic if and only if they are of the same dimension.

Solution:

$\rightarrow$: Trivial since isomorphism keeps bases distinct (so a basis of n elements maps to another basis of n elements).

$\leftarrow$: Consider any isomorphism of the underlying vector spaces (this exists because all complex vector spaces of the same dimension are isomorphic) and call it ϕ . For any $x, y \in L_1$, $0 = \phi(0) = \phi([x, y]) = [\phi(x), \phi(y)] = 0$, so the bracket is preserved.

Problem 3.1.4. (4 points) Assume that for all $x, y, z \in L$, $[x, [y, z]] = [[x, y], z]$. What is the most general statement you can make about elements of L ?

Solution: $[L, L] \subseteq Z(L)$ ($Z(L)$ is L 's center), or, put less symbolically, anything bracketed with anything else commutes with anything in L . Another way to put it is that

$[L, [L, L]] = 0$, or anything that uses that but with generic elements of L . This is due to the Jacobi identity: $0 = [x, [y, z]] + [y, [z, x]] + [z, [x, y]] = [y, [z, x]]$ by cancellation; since this is the case for any x, y, z , $[L, L] \subseteq Z(L)$ is all we can say.

Problem 3.1.5. Fix n . Suppose there is some subalgebra $S \subseteq \mathfrak{gl}_n$ for which $\phi : \mathfrak{sl}_2 \rightarrow S$ is a bijective Lie algebra homomorphism (where $\mathfrak{gl}_n$ has the bracket as in Problem 3.1.1). If $v_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $v_2 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, let $V_i = \phi(v_i)$. Let λ_3 and λ'_3 be the greatest and least eigenvalue, respectively, of V_3 and let $v, v' \in \mathbb{C}^n$ be vectors such that $V_3 v = \lambda_3 v$ and $V_3 v' = \lambda'_3 v'$.

- (i) (5 points) Find, with proof, $V_1 v$ and $V_2 v'$.
- (ii) (7 points) Find, with proof, λ_3 and λ'_3 .

Solution:

- (i) We first compute $[v_3, v_1] = 2v_1$ and $[v_3, v_2] = -2v_2$. $\phi([v_3, v_1])v = [\phi(v_3), \phi(v_1)]v$ and expanding gets $V_3 V_1 v = (\lambda + 2)V_1 v$. But we assumed λ was the maximal real eigenvalue – this is a contradiction if $V_1 v \neq 0$. Thus we conclude that it is indeed 0. By analogous computation, $V_3 V_2 v = (\lambda - 2)V_2 v$, and $V_3 V_2 v' = (\lambda' - 2)V_2 v'$, which forces $V_2 v' = 0$.
- (ii) We repeat the process of multiplying $V_3 V_2 v$ to get a string of eigenvectors ($v = v_\lambda, v_{\lambda-2}, \dots, v_{-\lambda+2}, v_{-\lambda}$). We know that the string must stop here because each is an eigenvector of V_3 and $\text{tr} V_3 = \text{tr} \phi(v_3) = \text{tr}(\phi([v_1, v_2])) = \text{tr}(V_1 V_2 - V_2 V_1) = 0$.

$$\sum_{k=0}^n (\lambda - 2k) = 0$$

$$(n+1)\lambda - n(n+1) = 0$$

$$\lambda = n$$

and so the minimum must be $\lambda' = -n$.

Problem 3.1.6. (4 points) Up to isomorphism, how many Lie algebras have dimension 1? dimension 2? List them all.

Solution: Dimension 1: any such Lie algebra is abelian, since its elements take the form ax for some nonzero vector x , and thus $[ax, bx] = ab[x, x] = 0$. This is unique up to isomorphism by Problem 3.1.3.

Dimension 2: If it's abelian then we've found the only one up to isomorphism, again by Problem 3.1.3. If it's not abelian, then we know that the set of post-bracket operations (e.g. $[v, w]$ for $v, w \in L$) is spanned by $[e, f]$ for e, f the bases of L . (This is because all elements are of the form $ae + bf$, so $[ae + bf, ce + df] = (ad - bc)[e, f]$.) If $[e, f] = 0$ then this is an abelian algebra. We actually find that $[e, f] = x$ for some $v \neq 0$. We can actually take a change-of-basis isomorphism to make x one of the basis elements and y the other. Thus up to isomorphism the last option is given by $[x, y] = x$.

3.2 Ideals and Subalgebras (16 points)

Problem 3.2.1.

- (i) (1 point) If L is an abelian Lie algebra, show that *any* subspace of L is a subalgebra and an ideal.

Solution: Let $V \subset L$ be a vector space. If $v \in V, x \in L$ then $[v, x] = 0 \in V$, so V is an ideal and thus a subalgebra.

- (ii) (4 points) Find a nonabelian Lie algebra on a space of $n \times n$ matrices for some $n > 2$ that has a subalgebra that is not an ideal. Specify that subalgebra and briefly justify why it is not an ideal.

Solution: Let $L = \mathfrak{gl}_3$ and let $S \subset L$ be the subalgebra of diagonal matrices. S is a subalgebra because it is a vector space with bracket identically 0. However, consider $x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} \in S$ and $y = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \in L \setminus S$. However, $[x, y] = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix} \notin S$.

Problem 3.2.2. (5 points) Take $I = \{x \in L \mid x \text{ nilpotent}\}$ where L is a subspace of $\mathfrak{gl}_n$. Is I a subalgebra? an ideal? Fully justify your responses.

Solution: The intuitive reasoning is that either I is a subalgebra but not an ideal, or it's both. As it turns out, however, neither is the case. Consider $N_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, N_2 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$.

$N_1, N_2 \in I$. $[N_1, N_2] = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ which is never 0 (its exponents have period 2). This extends to $N_i = \begin{bmatrix} 0 & \cdots & 2-i \\ \vdots & \ddots & \vdots \\ i-1 & \cdots & 0 \end{bmatrix}$ where all entries are zero except in the corners indicated;

here, the general case, $[N_1, N_2] \notin I$ for any value of n , making I not even closed under the Lie bracket.

Problem 3.2.3. (3 points) Let $[L_1, L_2] = \{[v_1, v_2] \mid v_i \in L_i\}$ for Lie algebras L_1 and L_2 . Is $[I, J]$ an ideal for ideals I and J of the same Lie algebra?

Solution: Yes. Take $a \in [I, J], z \in L$. We wish to have $[a, z] \in [I, J]$. a is of the form $[x, y]$ for some $x \in I, y \in J$. The Jacobi identity gives that $[[x, y], z] = -[[y, z], x] - [[z, x], y]$. $[y, z] \in J$ and $[z, x] \in I$, so indeed $[[x, y], z] \in [I, J]$.

Problem 3.2.4. (3 points) Show that $\ker \phi$ is an ideal and $\text{im } \phi$ is a subalgebra of the domain and image space, respectively, for ϕ a Lie algebra homomorphism.

Solution: If $x \in \ker \phi$ then $\phi(x) = 0$. $[y, x] \in \ker \phi$ iff $\phi([y, x]) = 0$. $\phi([y, x]) = [\phi(y), \phi(x)] = [\phi(y), 0] = 0$, as desired. The Jacobi identity can also be shown. For the image, we can show the Jacobi identity because any three elements appear as $\phi(x), \phi(y), \phi(z)$ for some x, y, z in the domain space; then, $[\phi(x), [\phi(y), \phi(z)]] = [\phi(x), \phi([y, z])] = \phi([x, [y, z]])$; summing cyclically gets what we want. The other properties are quicker to check.

3.3 Ado's Theorem (9 points)

Problem 3.3.1. Consider the additive vector space V_d of degree- d polynomials with real coefficients.

- (i) (1 point) Find the dimension of V_3 , and find a basis.

Solution: V_3 is polynomials of the form $a + bx + cx^2 + dx^3$, so the basis could be $\{1, x, x^2, x^3\}$ and the dimension is 4.

- (ii) (5 points) Say V_2 is given a bracket operation, for $p(x) = \alpha_0 + \alpha_1 x + \alpha_2 x^2$ and $q(x) = \beta_0 + \beta_1 x + \beta_2 x^2$ elements of V_2 ,

$$[p, q] = (\alpha_1 \beta_2 - \alpha_2 \beta_1) + (\alpha_2 \beta_0 - \alpha_0 \beta_2)x + (\alpha_0 \beta_1 - \alpha_1 \beta_0)x^2.$$

Find a subspace S of $\mathfrak{gl}_n$ for some n such that there is an isomorphism between V_2 and S ; explicitly describe such an isomorphism that preserves the bracket.

Solution: One possibility: $n = 3$. $\phi : p \mapsto \begin{bmatrix} 0 & -\alpha_2 & \alpha_1 \\ \alpha_2 & 0 & -\alpha_0 \\ -\alpha_1 & \alpha_0 & 0 \end{bmatrix}$, and

$$V_2 \stackrel{\phi}{\cong} \text{span} \left\{ \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \right\}.$$

Problem 3.3.2. (3 points) Suppose $\mathbb{R}^2$ is given a Lie bracket that makes it abelian. Find a subspace S of $\mathfrak{gl}_n$ for some n such that there is an isomorphism between $\mathbb{R}^2$ and S ; explicitly describe the isomorphism.

Solution: One possibility: $n = 2$. $\phi : (a, b) \mapsto \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$, and $\mathbb{R}^2 \stackrel{\phi}{\cong} \text{span} \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$.

3.4 The adjoint representation (26 points)

Problem 3.4.1. (1 point) Verify that $\text{ad } x$ is a linear map.

Solution: For any $z \in L$, $(\text{ad}(ax + by))(z) = [ax + by, z] = a[x, z] + b[y, z] = a(\text{ad } x)z + b(\text{ad } y)z$.

Problem 3.4.2.

- (i) (2 points) Explain how $\text{ad } x$ can be thought of as an element of $\mathfrak{gl}_3$ if $x \in \mathfrak{sl}_2$.

Solution: $\mathfrak{sl}_2$ is a 3-dimensional algebra, which means that it has 3 basis elements; $\text{ad } x : L \rightarrow L$, so to write it as a matrix we would consider how the basis elements are transformed under the function.

- (ii) (5 points) Find an explicit isomorphism from the Lie algebra $\mathfrak{sl}_2$ to a subalgebra of $\mathfrak{gl}_3$.

Solution: $[v_1, v_1] = 0, [v_1, v_2] = v_3, [v_1, v_1] = -2v_1$, so $\text{ad } v_1 = \begin{bmatrix} 0 & 0 & -2 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$. Simi-

lar computation gives $\text{ad } v_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 2 \\ -1 & 0 & 0 \end{bmatrix}$ and $\text{ad } v_3 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. We then find

$\text{ad } x \in \mathfrak{gl}_3$ by finding $x = ae_1 + be_2 + ce_3$ and setting $\text{ad } x = a\text{ad } e_1 + b\text{ad } e_2 + c\text{ad } e_3$.

Problem 3.4.3. (5 points) Show that the set of 3×3 complex upper-triangular matrices is a Lie algebra, find a basis, and write $\text{ad } e_i$ explicitly for each basis vector e_i . Find $\text{ad } e_i$'s eigenvalues for each basis vector e_i .

Solution: L has the basis $e_1 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, e_3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. $[e_1, e_1] =$

$0, [e_1, e_2] = 0, [e_1, e_3] = e_2$, so $\text{ad } e_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$. Similar computation gives $\text{ad } e_2 = 0$ and

$\text{ad } e_3 = \begin{bmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. These were computed by considering how each basis vector transforms under action by each adjoint.

Problem 3.4.4. (5 points) Show that $[x, y] = 0$ implies $(\operatorname{ad} x) \circ (\operatorname{ad} y) = (\operatorname{ad} y) \circ (\operatorname{ad} x)$.

Solution: Consider taking these functions on any $z \in L$. $(\operatorname{ad} x \circ \operatorname{ad} y)(z) = \operatorname{ad} x([y, z]) = [x, [y, z]] = -[y, [z, x]] - [z, [x, y]] = -[y, [z, x]] = [y, [x, z]] = \operatorname{ad} y \circ (\operatorname{ad} x)(z) = (\operatorname{ad} y \circ \operatorname{ad} x)(z)$. Facts used: Jacobi identity, anticommutativity.

Problem 3.4.5. (6 points) If x is nilpotent, is $\operatorname{ad} x$ nilpotent? If $\operatorname{ad} x$ is nilpotent, is x nilpotent? Justify your response.

Solution: For the first question: yes. Say $x^k = 0$. Then for any y take $(\operatorname{ad} x)^{2k}y = \sum_{m=0}^{2k} (-1)^{f(m)} x^m y^{2k-m} x^{2k-m}$ for some function f of m that is immaterial, since either $m \geq k$ for $2k - m \geq k$, so all terms go to zero, regardless of y . For the second question: no, $(\operatorname{ad} I_n)y = 0$ for any y , making $\operatorname{ad} I_n$ trivially nilpotent, but $I_n^k = I_n$ for any k .

Problem 3.4.6. (2 points) Show that $\operatorname{ad} x$ satisfies the relation $(\operatorname{ad} x)([y, z]) = [(\operatorname{ad} x)(y), z] + [y, (\operatorname{ad} x)(z)]$ for any $x, y, z \in L$. (Such functions are called *derivations*.)

Solution: $(\operatorname{ad} x)([y, z]) = [x, [y, z]] = -[y, [z, x]] - [z, [x, y]] = [y, [x, z]] + [[x, y], z] = [y, (\operatorname{ad} x)(z)] + [(\operatorname{ad} x)(y), z]$.

4 Lie Algebras II (55 points)

4.1 The Killing form and semisimple Lie algebras (24 points)

Problem 4.1.1. (1 point) Verify that the Killing form is symmetric and bilinear.

Solution: Linearity: ad is linear so $\kappa(ax, by) = \operatorname{tr}(\operatorname{ad}(ax) \circ \operatorname{ad}(by)) = \operatorname{tr}(ab \operatorname{ad} x \circ \operatorname{ad} y) = ab \operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad} y)$. $\kappa(x, y + z) = \operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad}(y + z)) = \operatorname{tr}(\operatorname{ad} x \circ (\operatorname{ad} y + \operatorname{ad} z)) = \operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad} y + \operatorname{ad} x \circ \operatorname{ad} z) = \kappa(x, y) + \kappa(x, z)$. The analogue holds for the other component.

Symmetry: $\kappa(x, y) = \operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad} y) = \operatorname{tr}(\operatorname{ad} y \circ \operatorname{ad} x) = \kappa(y, z)$.

Facts used: $\operatorname{tr}(XY) = \operatorname{tr}(YX)$, linearity of tr .

Problem 4.1.2. (5 points) Show that $\kappa([x, y], z) = \kappa(x, [y, z])$.

Solution:

$$\begin{aligned} \kappa([x, y], z) &= \kappa(xy - yx, z) = \operatorname{tr}(\operatorname{ad}(xy - yx) \operatorname{ad} z) \\ &= \operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad} y \circ \operatorname{ad} z - \operatorname{ad} y \circ \operatorname{ad} x \circ \operatorname{ad} z) \\ &= \operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad} y \circ \operatorname{ad} z) - \operatorname{tr}(\operatorname{ad} y \circ \operatorname{ad} x \circ \operatorname{ad} z) \\ &= \operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad} y \circ \operatorname{ad} z) - \operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad} z \circ \operatorname{ad} y) \\ &= \operatorname{tr}(\operatorname{ad} x \circ \operatorname{ad} y \circ \operatorname{ad} z - \operatorname{ad} x \circ \operatorname{ad} z \circ \operatorname{ad} y) \\ &= \operatorname{tr}(\operatorname{ad} x \circ (\operatorname{ad} y \circ \operatorname{ad} z - \operatorname{ad} z \circ \operatorname{ad} y)) \\ &= \kappa(x, [y, z]). \end{aligned}$$

Facts used: $\operatorname{tr}(XY) = \operatorname{tr}(YX)$, linearity of tr/κ .

Problem 4.1.3. (10 points) Show that $x \in L$ is semisimple if and only if $\operatorname{ad} x$ is semisimple.

Solution: We note that ad is injective, because if $\operatorname{ad} x = \operatorname{ad} y$ then $\operatorname{ad}(x - y) = 0$, but then $\kappa(x - y, z) = 0$ for all $z \in L$, so $x - y = 0$ by nondegeneracy. Any element of a matrix space can be uniquely written as the sum of a semisimple and nilpotent component (where those components commute, though this won't be necessary for us right now). Thus if we can show that x semisimple implies $\operatorname{ad} x$ semisimple we'll be done, since the inverse of the

$\text{ad } x$ map will take a semisimple adjoint plus a zero nilpotent part back to some other part plus a zero nilpotent part in L . Suppose x has eigenvectors e_i with eigenvalue λ_i . We may give $\text{ad } L$'s matrix space a subbasis of $e_{i,j}$, the $\dim L \times \dim L$ matrix with 1 at row i and column j . Computation gives that $(\text{ad } x)e_{i,j} = (\lambda_i - \lambda_j)e_{i,j}$, so the sum of $\text{ad } x$'s eigenspace's dimensions is maximal, since that is equivalent to being diagonalizable (since the structure of eigenspaces is preserved by similarity). Therefore we see that since x semisimple implies $\text{ad } x$ semisimple, we are done.

Problem 4.1.4.

- (i) (3 points) Let $I' = \{x \in L \mid \kappa(x, y) = 0 \forall y \in I\}$ for $I \subseteq L$ an ideal. Show that I' is also an ideal.
- (ii) (5 points) If L is semisimple, show that there are n simple subalgebras L_i ($1 \leq n \leq \dim L$) such that

$$L = \bigoplus_{i=1}^n L_i.$$

Solution:

- (i) Take any $z \in L$. If $[x, z] \in I'$ for $x \in I'$ then $\kappa([x, z], y) = 0$ for any $y \in I$. We manipulate κ as in Problem 4.1.2 to find $\kappa([x, z], y) = \kappa(x, [z, y])$. $[z, y] \in I$ since it is an ideal so we find that $\kappa([x, z], y) = \kappa(x, [z, y]) = 0$ as desired.
- (ii) We actually will show this for ideals. Let $I \subset L$ have least nonzero dimension of all ideals. (If $I = L$ then we're finished.) We may write $L = I \oplus I'$ since $I \cap I'$ is trivial. By dimension additivity, $\dim I' < \dim L$, so we descend by continuing this process on I' .

4.2 Root space decomposition (31 points)

Problem 4.2.1. (5 points) Show that each λ_i is linear.

Solution: $(\alpha\lambda(h_1) + \beta\lambda(h_2))v = [\alpha h_1 + \beta h_2, v] = \lambda(\alpha h_1 + \beta h_2)v$ as desired.

Problem 4.2.2. (3 points) Show that L is isomorphic to the vector space

$$\bigoplus_{\lambda: H \rightarrow \mathbb{C}} L_\lambda.$$

Solution: Each λ is a linear map, so L_λ contains $\text{span}\{v\}$, the λ -eigenvectors. In addition, each vector in L belongs to one such eigenspace. To see this, note that L has a basis of simultaneous eigenvectors of H , which is how we know that the λ s exist. If $v_1 \in L_\alpha$ and $v_2 \in L_\beta$ then $[h, v_1 + v_2] = [h, v_1] + [h, v_2] = \alpha(h)v_1 + \beta(h)v_2 \neq \gamma(h)(v_1 + v_2)$ for some $\gamma \in H^*$ only when $\alpha(h) = \beta(h)$ for all $h \in H$. We therefore find that when $\alpha \neq \beta$ these sets never have nonzero overlap, and thus they are linearly independent from each other, and we can add the spaces in the desired manner.

Problem 4.2.3. (6 points) Suppose α and β are roots of L having maximal toral subalgebra H . Under what condition on α and β is it true that $\kappa(v_1, v_2) \neq 0$ for some $v_1 \in L_\alpha, v_2 \in L_\beta$?

Solution: Restatement: if [some condition], then $\kappa(v_1, v_2) = 0$. We see that when the condition is $\alpha + \beta \neq 0$, we can provide the following argument: by nondegeneracy of $\alpha + \beta$ there is $h \in H$ with $(\alpha + \beta)(h) \neq 0$. Then, $\alpha(h)\kappa(v_1, v_2) = \kappa(\alpha(h)v_1, v_2) = \kappa([h, v_1], v_2) = -\kappa([v_1, h], v_2) = -\kappa(v_1, [h, v_2]) = -\kappa(v_1, \beta(h)v_2) = -\beta(h)\kappa(v_1, v_2)$. Adding gets $(\alpha + \beta)(h)\kappa(v_1, v_2) = 0$, and we are done.

Problem 4.2.4. (6 points) Show that $\langle \cdot, \cdot \rangle : H^* \times H^* \rightarrow \mathbb{C}$ as defined above is an inner product.

Solution: We know that this inner product is a symmetric bilinear form via what we know about κ . We need to check that $\langle \lambda, \lambda \rangle$ is always positive and real for nonzero $\lambda \in H^*$. $\langle \lambda, \lambda \rangle = \kappa(t_\lambda, t_\lambda) = \sum_{\alpha \in R_H} \alpha(t_\lambda)^2 = \sum_{\alpha \in R_H} \kappa(t_\alpha, t_\lambda)^2 = \sum_{\alpha \in R_H} \langle \alpha, \lambda \rangle^2 \in \mathbb{R}^+$. (R_H is the set of roots in H^* .) This follows from several results: H has simultaneous eigenvectors; the definition of the inner product on H^* ; and the fact that the inner product is rational-values on R_H since κ is the trace of matrices with integer eigenvalues.

Problem 4.2.5. (6 points) Show that $[x, y] = \kappa(x, y)t_\lambda$ for $x \in L_\lambda, y \in L_{-\lambda}$.

Solution: For any $h \in H$, $\kappa(h, [x, y]) = \kappa([h, x], y) = \lambda(h)\kappa(x, y) = \kappa(t_\lambda, h)\kappa(x, y) = \kappa(h, t_\lambda\kappa(x, y))$. Subtracting gives $0 = \kappa(h, [x, y] - \kappa(x, y)t_\lambda)$ for any h . The second component is independent of the first so we find that, independent of h , $[x, y] = \kappa(x, y)t_\lambda$ as desired.

Problem 4.2.6. (5 points) Show that for $\lambda \in H^*$, $\kappa(t_\lambda, t_\lambda)\kappa(h_\lambda, h_\lambda) = 4$ and show that $h_\lambda = \frac{2t_\lambda}{\kappa(t_\lambda, t_\lambda)}$.

Solution: $\lambda(h_\lambda) = 2$ by construction. Thus also by construction

$$2 = \lambda(h_\lambda) = \kappa(t_\alpha, h_\lambda) = \kappa(t_\alpha, [e_\lambda, f_\lambda]) = \kappa(t_\lambda, \kappa(e_\lambda, f_\lambda)t_\lambda) = \kappa(t_\lambda, t_\lambda)\kappa(e_\lambda, f_\lambda).$$

As $h_\lambda = \kappa(e_\lambda, f_\lambda)$, we substitute $\kappa(t_\lambda, t_\lambda)\kappa(e_\lambda, f_\lambda) = 2$ to find that $h_\lambda = \frac{2t_\lambda}{\kappa(t_\lambda, t_\lambda)}$, the second desired result. We use this to directly derive the first desired result: $\kappa(h_\lambda, h_\lambda) = \kappa\left(\frac{2t_\lambda}{\kappa(t_\lambda, t_\lambda)}, \frac{2t_\lambda}{\kappa(t_\lambda, t_\lambda)}\right) = \frac{4\kappa(t_\lambda, t_\lambda)}{\kappa(t_\lambda, t_\lambda)^2} = \frac{4}{\kappa(t_\lambda, t_\lambda)}$ which rearranges to the first desired result.

5 Root systems (125 points)

5.1 What is a root system? (37 points)

Problem 5.1.1. (1 point) Find all root systems in $\mathbb{R}$.

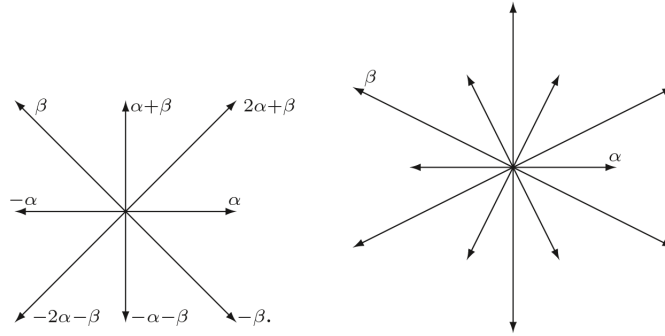
Solution: We know that a root system must span its vector space, and contain only two scalar multiples of any given element. Thus all possible root systems are $\{r, -r\}$ for any given $r \in \mathbb{R}^+$.

Problem 5.1.2. (3 points) In condition 3, note that v and w are interchangeable. Use this and the identity that $\langle v, w \rangle = \|v\| \cdot \|w\| \cos \phi$ (where ϕ is the angle between the vectors) to show that $4 \cos^2 \phi \in \{0, 1, 2, 3\}$ whenever $v \neq \pm w$.

Solution: $\langle v, w \rangle \langle w, v \rangle = 4 \left(\frac{\langle v, w \rangle}{\langle v, v \rangle} \right) \left(\frac{\langle w, v \rangle}{\langle w, w \rangle} \right) = 4 \cos^2 \phi$. v and w are in R only if $\langle v, w \rangle \in \mathbb{Z}$. Thus we find that $4 \cos^2 \phi$ is a positive integer no larger than 3.

Problem 5.1.3. (8 points) Think about the construction of the root systems in the text, as well as Problem 5.1.2, use these to understand why the subsets of $\mathbb{R}^2$ shown in Figure 1 are root systems. Draw (on separate coordinate axes) all possible root systems R .

Solution: There are 4 total possible root systems in $\mathbb{R}^2$. Two were mentioned in the contest. The following are B_2 and G_2 , respectively:



In the former, $\alpha = (1, 0)$ and $\beta = (-1, 1)$. In the latter, $\alpha = (1, 0)$ and $\beta = \left(-\frac{3}{2}, \frac{\sqrt{3}}{2}\right)$.

Problem 5.1.4. (10 points) Show that any root system can be written as the union of irreducible root systems. (Use the fact that a root system R is irreducible if there is no subset R' such that for $\alpha \in R'$ and $\beta \in R \setminus R'$, $\langle \alpha, \beta \rangle = 0$.)

Solution: Let $a \equiv b$ when there is a path of edges in the Dynkin diagram from the node corresponding to a to that of b . Let R_i enumerate the equivalence classes from this relation. Each of the R_i is a root system, by checking the root system properties, and necessarily they are each irreducible, since an individual R_i may not be split into mutually-orthogonal sets, since otherwise there would be $S \subset R_i$ for which $\langle s, r \rangle = 0$ for each $s \in S, r \in R_i \setminus S$, which would contradict the accessibility all elements from all other elements in R_i .

Problem 5.1.5. (15 points) Show that any root system has a base.

Solution: Clearly if $\dim V = 1$ there is a base $\{1\}$, for instance, suffices. Now, say $\dim V \geq 2$. Let $R^\perp = \{x \in V \mid (x, v) = 0 \forall v \in R\}$ (this is a nonempty set because it requires an uncountable number of $(n-1)$ -dimensional objects to fully cover a n -dimensional space). Fix some $x \in R^\perp$. Let $R^+ = \{v \in R \mid (x, v) > 0\}$. Let B be the elements of R^+ that are not the sum of two other elements of R^+ .

Suppose B was not linearly independent. Then there are real values r_v for which $\sum_{v \in B} r_v v = 0$.

Let B^+ be the elements with $r_v > 0$ and B^- have $r_v < 0$. Set $x' = \sum_{B^+} r_v v = \sum_{B^-} (-r_v) v$. Now we take the inner product of these two formulations of x' . $\langle x', x' \rangle = \langle \sum_{B^+} r_v v, \sum_{B^-} (-r_v) v \rangle = \sum_{v_+ \in B^+, v_- \in B^-} r_{v_+} (-r_{v_-}) \langle v_+, v_- \rangle$. We note that the inner prod-

uct of all pairs of distinct base elements must be negative, i.e. they form obtuse angles; this is because $|B| = \dim V$, and we must satisfy the restrictions on possible angle cosines (which may only have size $0, \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, 1$, so we are constrained in our choice of angles). However, $\langle x', x' \rangle \geq 0$ and this is the sum of positive quantities multiplied by negative inner products, which is negative. Thus we find that $\langle x', x' \rangle = 0$, i.e. $x' = 0$. Taking $\langle x', x \rangle$ gets $0 = \langle 0, x \rangle = \sum_{B^+} r_v \langle v, x \rangle$; $r_v, \langle v, x \rangle \geq 0$ with strict inequality for the inner product, so we find that $r_v = 0$ for all $v \in B$. This contradicts B 's linear dependence. Additionally, either v or $-v$ is in B (since $\langle v, x \rangle > 0$ or $\langle v, x \rangle < 0$), so $|B| = \dim V$.

We must also show that B satisfies the other base property. Take any $w \in R$, which lies in either R^+ or $-w \in R^+$. WLOG the former. Suppose there doesn't exist coefficients k_v for which $\sum_{B} k_v v = w$. Then, choose the element $w \in R^+$ for which $\langle x, w \rangle$ is minimized. $w \notin B$ (otherwise we'd've chosen $k_w = 1$ and the others to be 0), so there are $w_1, w_2 \in R^+$ with $w = w_1 + w_2$. $\langle x, w \rangle = \langle x, w_1 \rangle + \langle x, w_2 \rangle$, both of which are positive, so $0 < \langle x, w_1 \rangle, \langle x, w_2 \rangle < \langle x, w \rangle$ so then w_1, w_2 or both may not be written in the form of the second condition. This contradicts the minimality of $\langle x, w \rangle$.

5.2 Dynkin diagrams (48 points)

Problem 5.2.1. (10 points) Show that a root system is irreducible if and only if it has a connected Dynkin diagram.

Solution:

→: If a root system is irreducible, then it has a connected Dynkin diagram. Suppose not; that there was an irreducible root system with a disconnected diagram. Let R_1 be one of the disjoint components. Then, there are no edges from any of R_1 to the rest of the root system...but we have defined the edge count between v and w to be in nonzero proportion with $\langle v, w \rangle^2$, which means that $\langle v, w \rangle = 0$ for $v \in R_1, w \in R \setminus R_1$. This contradicts irreducibility.

←: If a Dynkin diagram is connected, then its root system is irreducible. Again we suppose not, and that it is reducible. Then there would be a subset $R_1 \subset R$ for which $\langle v, w \rangle = 0$ when $v \in R_1, w \in R \setminus R_1$. However, the vertices in the Dynkin diagram corresponding to these vectors are connected to at least something outside of R_1 by connectedness. Again, we have defined the edge count between v and w to be in nonzero proportion with $\langle v, w \rangle^2$, which means that $\langle v, w \rangle \neq 0$ for some $v \in R_1, w \in R \setminus R_1$. This contradicts reducibility.

Problem 5.2.2. (7 points) Show that no Dynkin diagram may have a cycle.

Solution: We actually show a stronger result: that no set that we can turn into a Dynkin diagram may have as many vertices as edges, as a cycle would (since it'd be a sub-diagram).

Suppose the vectors $\{v_1, \dots, v_n\}$ can form a valid Dynkin diagram. Then if $v = \sum v_i$, $0 < \langle v, v \rangle = n + 2 \sum_{i < j} \langle v_i, v_j \rangle$ (since the v_i 's are unit vectors). So, $n > \left| \sum_{i < j} 2 \langle v_i, v_j \rangle \right| \geq C$

since whenever v_i and v_j are connected they have $4 \langle v_i, v_j \rangle^2 \in \{1, 2, 3\}$, and where C is the number of such pairs. Thus we find that $n > C$, as desired, and the proof is finished.

Problem 5.2.3. (8 points) Show that no node in a Dynkin diagram may have more than three total edges connected to it.

Solution: Say v is connected to $v_1, \dots, v_n$. There are no cycles, so $\langle v_i, v_j \rangle = 0$ when $i \neq j$. We may find v_0 with $\langle v, v_0 \rangle \neq 0$ such that $\text{span}\{v_0, \dots, v_n\} = \text{span}\{v, v_1, \dots, v_n\}$. Thus we may write $v = \sum \langle v, v_i \rangle v_i$ and then taking the inner product with v , since v is a unit vector, gives $1 = \langle v, v \rangle = \sum \langle v, v_i \rangle^2$. Since $0 < \langle v, v_0 \rangle < 1$, we find $\sum \langle v, v_i \rangle^2 < 1$. Since $4 \langle v, v_i \rangle^2 \in \{1, 2, 3\}$, $\langle v, v_i \rangle \geq \frac{1}{4}$ so we find that n may not be 4 or more.

Problem 5.2.4. (4 points) The *Shrinking Lemma* states that if a Dynkin diagram on the base B has n vertices $v_1, v_2, \dots, v_n$ that are in a *line* and $v = \sum_{i=1}^n v_i$, then the base $B \setminus \{v_1, v_2, \dots, v_n\} \cup \{v\}$ corresponds to the root system with the intermediate vertices “combined” into a single vertex (as before). (See Figure 4.) Use the Shrinking Lemma to show that no Dynkin diagram may have more than one branch, more than one double-edge, or both a branch and a double edge.

Solution: The first case is of two double edges. Some pair of them is connected by a line; invoke the Shrinking Lemma to turn that line into a single node, with two double edges attached to it. This violates Problem 5.2.3.

The second case is similar: there at least two branching points, with some pair of them connected by a line.

The third case follows for the same reason; we use the Shrinking Lemma to bring the branching point and double edge together, giving a quadruple-edge.



Figure 1: An impossible Dynkin diagram. The Shrinking Lemma gives that if there is a root system corresponding a variant of this diagram (with n nodes amongst the $\dots$) then there is one corresponding to the one with $n - 1, n - 2, \dots, 1, 0$ vertices.

Problem 5.2.5. (4 points) Prove that if a Dynkin Diagram has a double edge with both ends connected to other nodes then it must be F_4 .

Solution: Suppose there are m nodes to the left of the double-edge and n to the right (including the nodes that form the edge itself). Number them $v_1, \dots, v_m$ and $w_1, \dots, w_n$, respectively, where $4\langle v_m, w_n \rangle^2 = 2$ (and $\langle v_m, w_n \rangle^2 = \frac{1}{2}$), so the other v_i and w_i do not “touch” and thus $\langle v_i, w_j \rangle = 0$ otherwise. Additivity of inner products means that if we set $v = \sum v_i, w = \sum w_i$ then $\langle v, w \rangle^2 = \langle mv_m, nw_n \rangle^2 = m^2 n^2 \langle v_m, w_n \rangle^2 = \frac{1}{2} m^2 n^2$. We know that v and w are non-parallel vectors with some angle θ between them; $\langle v, w \rangle = \|v\| \|w\| \cos \theta < \|v\| \|w\|$, or $\langle v, w \rangle^2 < \|v\|^2 \|w\|^2 = \langle v, v \rangle \langle w, w \rangle$. We already know how to evaluate the left-hand side; this was done in Section 5.2. Substituting in what we know, $\frac{1}{2} m^2 n^2 < m(m+1)n(n+1)$, or $(m-1)(n-1) < 2$. We want $m, n > 1$ which means $m = n = 2$, as desired.

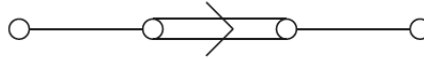


Figure 2: F_4 , the Dynkin diagram discussed in Problem 5.3.6.

Problem 5.2.6. (15 points) Imagine a Dynkin diagram with a “branching point” – there is some vector with three lines coming off of it. Show that one of those lines must have length one (i.e. only one edge), and show that of the other two, either

- if one has length 1 then the third may have any length
- if one has length 2 then the other’s length may not be more than 4.

Solution: We know that there are no double-edges, by Problem 5.2.4. Suppose the branching node is z , and it is joined to x_n, w_m, v_p where each of those are connected to a line going to 1 such as $x_{n-1}, \dots, x_1$. We create the “super-vertices” x, w and v as specified earlier. Each is orthogonal to the others. Normalize them to x', w', v' . We can give $\text{span}\{z, x, w, v\}$ the basis $\{z_0, x', w', v'\}$ for z_0 a unit vector not orthogonal to z but orthogonal to each of x', w', v' . It is rather readily seen that $z = \langle z, x' \rangle x' + \langle z, w' \rangle w' + \langle z, v' \rangle v' + \langle z, z_0 \rangle z_0$. z too is a unit vector (since it lies in a normalized base). Taking another inner product with z gives $1 = \langle z, x' \rangle^2 + \langle z, w' \rangle^2 + \langle z, v' \rangle^2 + \langle z, z_0 \rangle^2$, i.e. $\langle z, x' \rangle^2 + \langle z, w' \rangle^2 + \langle z, v' \rangle^2 < 1$ ($0 < \langle z, z_0 \rangle < 1$). We’ve already computed the inner products of x, w, v with themselves, and from the Dynkin diagram we know that $\langle z, nx_n \rangle = \frac{n^2}{4}$ (since they’re connected by a single edge). Substituting: $\frac{2n^2}{4n(n+1)} + \frac{2m^2}{4m(m+1)} + \frac{2p^2}{4p(p+1)} < 1$, and rearranging gets $\frac{1}{n+1} + \frac{1}{m+1} + \frac{1}{p+1} > 1$. Without loss of generality take $n \geq m \geq p$. Then $\frac{1}{n+1} \leq \frac{1}{m+1} \leq \frac{1}{p+1} \leq \frac{1}{2}$, or $1 < \frac{3}{p+1}$, so $p < 2$, i.e. $p = 1$. Then, $m < 3$, i.e. $m = 1, 2$. If $m = 2$ then $n < 5$; otherwise n can be any natural.

5.3 Dynkin diagrams of Lie algebras (40 points)

Problem 5.3.1. (10 points each) For $n \geq 1$ in all cases...

- draw the Dynkin diagram for $\mathfrak{so}_{2n+1}$

- (ii) draw the Dynkin diagram for $\mathfrak{sl}_n$
- (iii) draw the Dynkin diagram for $\mathfrak{sp}_{2n}$

and show your reasoning. In particular, many details of the computations were omitted above, but you will be expected to (briefly) justify the numbers you find.

Solution:

- (i) As L 's elements satisfy $x^T R_{2n} + R_{2n} x = 0$, we know that x can be written in the form

$\begin{bmatrix} 0 & C^T & -B^T \\ B & M & P \\ -C & Q & -M^T \end{bmatrix}$ where B and C each have n rows and 1 column and for any $n \times n$ matrix M and $n \times n$ matrices P and Q both equal to their negative transpose.

H is the set of diagonal matrices in L , so $H = \left\{ \sum_{i=1}^n c_i (e_{i,i} - e_{i+n,i+n}) \mid c_i \in \mathbb{C} \right\}$ (indexed from zero this time). We can give L the additional basis elements $m_{i,j} = e_{i,j} - e_{j+n,i+n}$ ($1 \leq i \neq j \leq n$), $p_{i,j} = e_{i,j+n} - e_{j,i+n}$ ($1 \leq i < j \leq n$), $p_{i,j}^T$, $B_i = e_{i,0} - e_{0,i+n}$ ($1 \leq i \leq n$) and $C_i = e_{0,i} - e_{i+n,0}$ ($1 \leq i \leq n$) to go along with those of H . We find, for an arbitrary element $h \in H$ of the form specified in H 's specification,

$$\begin{aligned} [h, m_{i,j}] &= (c_i - c_j) m_{i,j} \\ [h, p_{i,j}] &= (c_i + c_j) p_{i,j} \\ [h, p_{i,j}^T] &= -(c_i + c_j) p_{i,j}^T \\ [h, B_i] &= c_i B_i \\ [h, C_i] &= -c_i C_i \end{aligned}$$

so our roots are:

root	eigenspace
$\lambda_i - \lambda_j$	$\text{span} \{m_{i,j}, m_{j,i}\}$
$\lambda_i + \lambda_j$	$\text{span} \{p_{i,j}, p_{j,i}\}$
λ_i	$\text{span} \{B_i\}$
$-\lambda_i$	$\text{span} \{C_i\}$

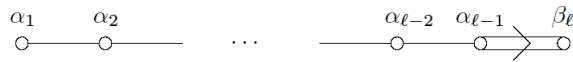
For H^* , a vector space having the basis $\{\lambda_i\}$, we see that $\{\lambda_1 - \lambda_2, \lambda_2 - \lambda_3, \dots, \lambda_{n-1} - \lambda_n, \lambda_n\}$ is a base; this base will be what we use to build our Dynkin diagram. We set $\alpha_i = \lambda_i - \lambda_{i+1}$ and $\beta = \lambda_n$. When $i < n$ if $e_{\alpha_i} = m_{i,i+1}$ then $h_{\alpha_i} = e_{i,i} - e_{i+n,i+n} - e_{i+1,i+1} + e_{i+n+1,i+n+1}$. $e_{\beta} = B_n$ gets $h_{\beta} = 2(e_{n,n} - e_{2n-2n})$. We compute

$$\begin{aligned} [h_{\alpha_j}, e_{\alpha_i}] &= \begin{cases} 2e_{\alpha_j} & i = j \\ -e_{\alpha_j} & |i - j| = 1 \\ 0 & \text{otherwise} \end{cases} \\ [h_{\beta}, e_{\alpha_i}] &= \begin{cases} -2e_{\alpha_i} & i = n - 1 \\ 0 & \text{otherwise} \end{cases} \\ [h_{\alpha_i}, e_{\beta}] &= \begin{cases} -e_{\beta} & i = n - 1 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

Computation gives that

$$\begin{aligned}\langle \lambda_i - \lambda_{i+1}, \lambda_j - \lambda_{j+1} \rangle &= \begin{cases} 2 & i = j \\ -1 & |i - j| = 1 \\ 0 & \text{otherwise} \end{cases} \\ \langle \lambda_i - \lambda_{i+1}, \lambda_n \rangle &= \begin{cases} -2 & i = n - 1 \\ 0 & \text{otherwise} \end{cases} \\ \langle \lambda_n, \lambda_i - \lambda_{i+1} \rangle &= \begin{cases} -1 & i = n - 1 \\ 0 & \text{otherwise} \end{cases}\end{aligned}$$

Since $e(v, w) = \langle v, w \rangle \cdot \langle w, v \rangle$, our Dynkin diagram is:



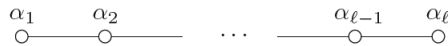
- (ii) H is the set of diagonal matrices in L , so $H = \left\{ \sum_{i=1}^{n-1} c_i(e_{i,i} - e_{n,n}) \mid c_i \in \mathbb{C} \right\}$. We can give L the additional basis elements $e_{i,j}$ ($1 \leq i \neq j \leq n$) to go along with those of H . We find, for an arbitrary element $h \in H$ of the form specified in H 's specification, $[h, e_{i,j}] = (c_i - c_j)e_{i,j}$, so our roots are $\lambda_i - \lambda_j$ with eigenspace $\text{span}\{e_{i,j}\}$. For H^* , a vector space having the basis $\{\lambda_i\}$, we see that $\{\lambda_1 - \lambda_2, \lambda_2 - \lambda_3, \dots, \lambda_{n-1} - \lambda_n\}$ is a base; this base will be what we use to build our Dynkin diagram. We set $\alpha_i = \lambda_i - \lambda_{i+1}$. When $i < n$ if $e_{\alpha_i} = e_{i,i+1}$ then $h_{\alpha_i} = e_{i,i} - e_{i+1,i+1}$. We compute

$$[h_{\alpha_j}, e_{\alpha_i}] = \begin{cases} 2e_{\alpha_j} & i = j \\ -e_{\alpha_j} & |i - j| = 1 \\ 0 & \text{otherwise} \end{cases}$$

Computation gives that

$$\langle \lambda_i - \lambda_{i+1}, \lambda_j - \lambda_{j+1} \rangle = \begin{cases} 2 & i = j \\ -1 & |i - j| = 1 \\ 0 & \text{otherwise} \end{cases}$$

Since $e(v, w) = \langle v, w \rangle \cdot \langle w, v \rangle$, our Dynkin diagram is where $\ell = n - 1$.



- (iii) As L 's elements satisfy $x^T \tilde{R}_{2n} + \tilde{R}_{2n} x = 0$, we know that x can be written in the form $\begin{bmatrix} M & P \\ Q & -M^T \end{bmatrix}$ for any $n \times n$ matrix M and $n \times n$ matrices P and Q both equal to their transposes.

H is the set of diagonal matrices in L , so $H = \left\{ \sum_{i=1}^n c_i(e_{i,i} - e_{i+n,i+n}) \mid c_i \in \mathbb{C} \right\}$. We can give L the additional basis elements $m_{i,j} = e_{i,j} - e_{j+n,i+n}$ ($1 \leq i \neq j \leq n$), $p_{i,j} = e_{i,j+n} + e_{j,i+n}$ ($1 \leq i < j \leq n$), $p_{i,j}^T$, $p_{i,i} = e_{i,i+n}$ ($1 \leq i \leq n$), and $p_{i,i}^T$ to go along with those of H . We find, for an arbitrary element $h \in H$ of the form specified in H 's specification,

$$\begin{aligned}[h, m_{i,j}] &= (c_i - c_j)m_{i,j} \\ [h, p_{i,j}] &= (c_i + c_j)p_{i,j} \\ [h, p_{i,j}^T] &= -(c_i + c_j)p_{i,j}^T\end{aligned}$$

(where in the latter two lines i may equal j) so our roots are:

root	eigenspace
$\lambda_i - \lambda_j$	$\text{span}\{m_{i,j}, m_{j,i}\}$
$\lambda_i + \lambda_j$	$\text{span}\{p_{i,j}, p_{j,i}\}$
$2\lambda_i$	$\text{span}\{p_{i,i}\}$
$-2\lambda_i$	$\text{span}\{p_{i,i}^T\}$

For H^* , a vector space having the basis $\{\lambda_i\}$, we see that $\{\lambda_1 - \lambda_2, \lambda_2 - \lambda_3, \dots, \lambda_{n-1} - \lambda_n, 2\lambda_n\}$ is a base; this base will be what we use to build our Dynkin diagram. We set $\alpha_i = \lambda_i - \lambda_{i+1}$ and $\beta = 2\lambda_n$. When $i < n$ if $e_{\alpha_i} = m_{i,i+1}$ then $h_{\alpha_i} = e_{i,i} - e_{i+n,i+n} - e_{i+1,i+1} + e_{i+n+1,i+n+1}$. $e_\beta = p_{n,n}$ gets $h_\beta = e_{n,n} - e_{2n,2n}$. We compute

$$[h_{\alpha_j}, e_{\alpha_i}] = \begin{cases} 2e_{\alpha_j} & i = j \\ -e_{\alpha_j} & |i - j| = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$[h_\beta, e_{\alpha_i}] = \begin{cases} -2e_{\alpha_i} & i = n - 1 \\ 0 & \text{otherwise} \end{cases}$$

$$[h_{\alpha_i}, e_\beta] = \begin{cases} -e_\beta & i = n - 1 \\ 0 & \text{otherwise} \end{cases}$$

Computation gives that

$$\langle \lambda_i - \lambda_{i+1}, \lambda_j - \lambda_{j+1} \rangle = \begin{cases} 2 & i = j \\ -1 & |i - j| = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\langle \lambda_i - \lambda_{i+1}, \lambda_n \rangle = \begin{cases} -2 & i = n - 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\langle \lambda_n, \lambda_i - \lambda_{i+1} \rangle = \begin{cases} -1 & i = n - 1 \\ 0 & \text{otherwise} \end{cases}$$

Since $e(v, w) = \langle v, w \rangle \cdot \langle w, v \rangle$, our Dynkin diagram is:



Problem 5.3.2. (10 points) State all isomorphisms between the classical Lie algebras, and the Dynkin diagrams that you used to reach those conclusions. (Do not provide the isomorphisms explicitly – just briefly state your reasoning.)

Solution: $\mathfrak{so}_3 \simeq \mathfrak{sp}_2 \simeq \mathfrak{sl}_2$ via root systems of type A1. $\mathfrak{so}_5 \simeq \mathfrak{sp}_4$ via root systems of types B2 and C2. $\mathfrak{so}_6 \simeq \mathfrak{sl}_4$ via root systems of types D3 and A3.